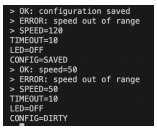
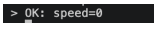
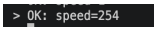
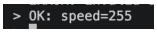
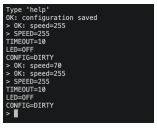
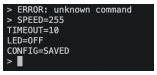
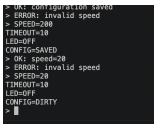
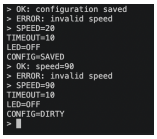
UART CLI Homework — Test Report

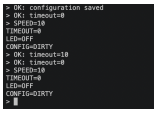
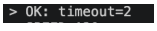
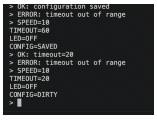
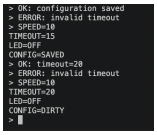
Student: Mariia Aliksieievych Date: October 4, 2026

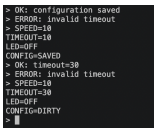
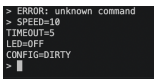
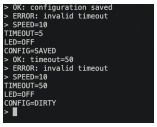
Objective: find robustness defects without inspecting the source code first.

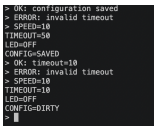
1. Test Execution

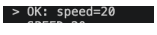
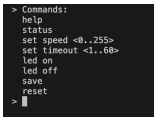
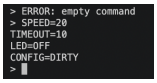
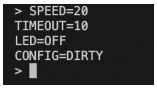
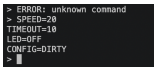
TC ID	Input / Sequence	Technique	Expected	Actual	Evidence	Result	Bug ID
TC-01	SAVED: save -> set speed -1 -> status DIRTY: set speed 50 -> set speed -1-> status	Boundary Value Analysis; State / Sequence Testing	'ERROR: speed out of range' in both states. Configuration unchanged; status remains available	'ERROR: speed out of range' in both states. Configuration unchanged; status remains available		PASS	
TC-02	set speed 0	Boundary Value Analysis	OK: speed=0	OK: speed=0		PASS	
TC-03	set speed 1	Boundary Value Analysis	OK: speed=1	OK: speed=1		PASS	
TC-04	set speed 254	Boundary Value Analysis	OK: speed=254	OK: speed=254		PASS	
TC-05	set speed 255	Boundary Value Analysis	OK: speed=255	OK: speed=255		PASS	
TC-06	SAVED: save -> set speed 256 -> status DIRTY: set speed 70 -> set speed 256 -> status	Boundary Value Analysis; State / Sequence Testing	'ERROR: speed out of range' in both states. Configuration unchanged; status remains available	OK: speed=255 in both states. Speed was changed instead of returning ERROR in both states.		FAIL	Bug-01
TC-07	Speed 1	Negative Testing	ERROR: unknown command; status remains available	ERROR: unknown command; status remains available		PASS	

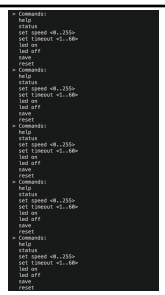
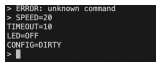
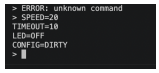
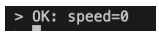
TC-08	SAVED: save -> set spe_ed 255 -> status DIRTY: set speed 50 -> set spe_ed 255 -> status	Negative Testing; State / Sequence Testing	‘ERROR: unknown command’ in both states. ; Configuration unchanged; status remains available	‘ERROR: unknown command’ in both states. ; Configuration unchanged; status remains available		PASS	
TC-09	SAVED: save -> set SPEED 254-> status DIRTY: set speed 70 ->set SPEED 254-> status	Negative Testing; State / Sequence Testing	‘ERROR: unknown command’ in both states. ; Configuration unchanged; status remains available	‘ERROR: unknown command’ in both states. ; Configuration unchanged; status remains available		PASS	
TC-10	SAVED: save -> set speed 2 78 -> status DIRTY: set speed 200 ->set speed 2 78 -> status	Negative Testing; State / Sequence Testing	‘ERROR: invalid speed’ in both states; Configuration unchanged; status remains available	‘ERROR: invalid speed’ in both states; Configuration unchanged; status remains available		PASS	
TC-11	SAVED: save -> set speed 🙈-> status DIRTY: set speed 20-> set speed 🙈-> status	Negative Testing; State / Sequence Testing	ERROR: invalid speed’ in both states; Configuration unchanged; status remains available	ERROR: invalid speed’ in both states; Configuration unchanged; status remains available		PASS	
TC-12	SAVED: save -> set speed 25,7-> status DIRTY: set speed 90 -> set speed 25,7-> status	Negative Testing; State / Sequence Testing	ERROR: invalid speed’ in both states; Configuration unchanged; status remains available	ERROR: invalid speed’ in both states; Configuration unchanged; status remains available		PASS	

TC-13	SAVED: save -> set speed -> status DIRTY: set speed 10 -> set speed -> status	Negative Testing; State / Sequence Testing	ERROR: invalid speed' in both states; Configuration unchanged; status remains available	ERROR: invalid speed' in both states; Configuration unchanged; status remains available		PASS	
TC-14	SAVED: save -> set timeout 0 -> status DIRTY: set timeout 10 -> set timeout 0 -> status	Boundary Value Analysis; State / Sequence Testing	'ERROR: timeout out of range' in both states. Configuration unchanged; status remains available	'OK: timeout=0' in both states. Timeout was changed instead of returning ERROR in both states.		FAIL	Bug-02
TC-15	set timeout 1	Boundary Value Analysis	OK: timeout=1	OK: timeout=1		PASS	
TC-16	set timeout 2	Boundary Value Analysis	OK: timeout=2	OK: timeout=2		PASS	
TC-17	set timeout 59	Boundary Value Analysis	OK: timeout=59	OK: timeout=59		PASS	
TC-18	set timeout 60	Boundary Value Analysis	OK: timeout=60	OK: timeout=60		PASS	
TC-19	SAVED: save -> set timeout 61 -> status DIRTY: set timeout 20 -> set timeout 61-> status	Boundary Value Analysis; State / Sequence Testing	'ERROR: timeout out of range' in both states. Configuration unchanged; status remains available	'ERROR: timeout out of range' in both states. Configuration unchanged; status remains available		PASS	
TC-20	SAVED: save -> set timeout zero -> status DIRTY: set timeout 20 -> set timeout	Negative Testing; Sequence Testing	'ERROR: invalid timeout' in both states. Configuration unchanged; status	'ERROR: invalid timeout' in both states. Configuration unchanged; status		PASS	

	zero -> status		remains available	remains available			
TC-21	SAVED: save -> set time_out 3 -> status DIRTY: set timeout 10 -> set time_out 3 -> status	Negative Testing; Sequence Testing	‘ERROR: unknown command’ in both states. Configuration unchanged; status remains available	‘ERROR: unknown command’ in both states. Configuration unchanged; status remains available		PASS	
TC-22	SAVED: save -> set timeout 🙈-> status DIRTY: set timeout 30 > set timeout 🙈-> status	Negative Testing; Sequence Testing	‘ERROR: invalid timeout’ in both states. Configuration unchanged; status remains available	‘ERROR: invalid timeout’ in both states. Configuration unchanged; status remains available		PASS	
TC-23	SAVED: save -> set timeout 25,5 -> status DIRTY: set timeout 5 -> set timeout 25,5 -> status	Negative Testing; Sequence Testing	‘ERROR: invalid timeout’ in both states. Configuration unchanged; status remains available	‘ERROR: invalid timeout’ in both states. Configuration unchanged; status remains available		PASS	
TC-24	Timeout	Negative Testing	ERROR: unknown command; status remains available	ERROR: unknown command; status remains available		PASS	
TC-25	SAVED: save -> set timeout 2 8 -> status DIRTY: set timeout 50 -> set	Negative Testing; Sequence Testing	‘ERROR: invalid timeout’ in both states. Configuration unchanged; status	‘ERROR: invalid timeout’ in both states. Configuration unchanged; status		PASS	

	timeout 2 8 -> status		remains available	remains available			
TC-26	SAVED: save -> set timeout: 7 -> status DIRTY: set timeout 10 -> set timeout: 7 -> status	Negative Testing; Sequence Testing	'ERROR: invalid timeout' in both states. Configuration unchanged; status remains available	'ERROR: invalid timeout' in both states. Configuration unchanged; status remains available		PASS	
TC-27	aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa

TC-29	set speed20 -> status	Negative Testing	Requirements Clarification Needed: Should space between the command and its argument be required?	OK: speed=20		?	
TC-30	help (space)	Negative Testing	Requirements Clarification Needed: Should trailing spaces after commands be ignored?	Help is displayed		?	
TC-31	Press Enter -> status	Negative Testing	ERROR: empty command; status remains available	ERROR: empty command; status remains available		PASS	
TC-32	STATUS	Negative Testing	ERROR: unknown command	Status is displayed		FAIL	BUG-03
TC-33	status help	Negative Testing	ERROR: unknown command; status remains available	ERROR: unknown command; status remains available		PASS	
TC-34	led on	Functional Testing	Requirements Clarification Needed. Exact response message is not specified.	OK		?	
TC-35	led off	Functional Testing	Requirements Clarification Needed. Exact response	OK		?	

			message is not specified.				
TC-36	Type help 5 times	Stress Input	Help is displayed 5 times	Help is displayed 5 times		PASS	
TC-37	abcdefghijklmnopqrstuvwxyz1234567890123 (39 symbols)	Input Length Testing	ERROR: unknown command; status remains available	ERROR: unknown command; status remains available		PASS	
TC-38	abcdefghijklmnopqrstuvwxyz12345678901234 (40 symbols)	Input Length Testing	ERROR: unknown command; status remains available	ERROR: unknown command; status remains available		PASS	
TC-39	abcdefghijklmnopqrstuvwxyz123456789012345 (41 symbols)	Input Length Testing	`ERROR: command too long`; status remains available	CRASH *** SYSTEM FAULT *** Reason: input line boundary failure rst:0xc (SW_CPU_RESET) Rebooting...		FAIL	BUG-04
TC-40	так	Negative testing	ERROR: unknown command	OK		FAIL	BUG-05
TC-41	set timeout 000	Negative testing	Requirements Clarification Needed: Are leading zeros allowed in numeric values?	OK: timeout=0		?	
TC-42	set speed 000	Negative testing	Requirements	OK: speed=0		?	

			Clarification Needed: Are leading zeros allowed in numeric values?				
TC-43	https://github.com/login	Negative testing	ERROR: unknown command; status remains available	ERROR: unknown command; status remains available		PASS	
TC-44	???	Negative testing	ERROR: unknown command	> OK		FAIL	BUG-05
TC-45	Enter set speed 9999999999999999995 times consecutively without Enter.	Stress Input	`ERROR: command too long`. speed remains unchanged; the device remains responsive and does not crash or reboot. Command should respond	CRASH. *** SYSTEM FAULT *** Reason: input line boundary failure rst:0xc (SW_CPU_RESET) Rebooting... After typing any command ('help', 'status'), it returns ERROR: unknown command		FAIL	BUG-06
TC-46	SAVE: set speed 50 -> save ->status -> set DIRTY: set speed 70 -> set	Stress Input	SAVE: 1.OK; speed=50 2.Status opens 3.ERROR: unknown command DIRTY: 1.OK; speed=70 2.ERROR: unknown command	SAVE: 1.OK; speed=50 2.Status opens DIRTY: 1.OK; speed=70 2. CRASH *** SYSTEM FAULT *** Reason: parser state		FAIL	BUG-07

				mismatch while configuration is dirty rst:0xc (SW_CPU_RESET) Rebooting...			
TC-47	Enter held -> press delete/back space -> Enter help	Negative testing	Requirements Clarification Needed: Should it be the ability to delete characters or clear the input line?	ERROR: unknown command		?	

2. Boundary / Length Analysis

Speed boundary values tested: -1, 0, 1, 254, 255, 256

Timeout boundary values tested: 0, 1, 2, 59, 60, 61

Input lengths tested: 39, 40, 41

Exact behavior-change boundary: 1) Speed: 256 is set to 255; 2) Timeout: 0 is accepted; 3) Input length: The system crashes at 41 characters.

Observed result at N-1 / N / N+1: 1) Speed: -1 - ERROR: speed out of range; 0 - OK: speed=0; 1 - OK: speed=1; 254 - OK: speed=254; 255 - OK: speed=255; 256 - OK: speed=255.

2) Timeout: 0 - OK: timeout=0; 1 - OK: timeout=1; 2 - OK: timeout=2; 59 - OK: timeout=59; 60 - OK: timeout=60; 61 - ERROR: timeout out of range.

3) Input lengths: 39 - ERROR: unknown command; 40 - ERROR: unknown command; 41 - CRASH.

3. Defects Found

BUG-01

Title: Speed was set to 255 after set speed 256 instead of returning an ERROR.

Steps:

1. Type command ‘save’
2. set speed 256
3. Type command ‘status’

4. set speed 70
5. set speed 256
6. Type command ‘status’

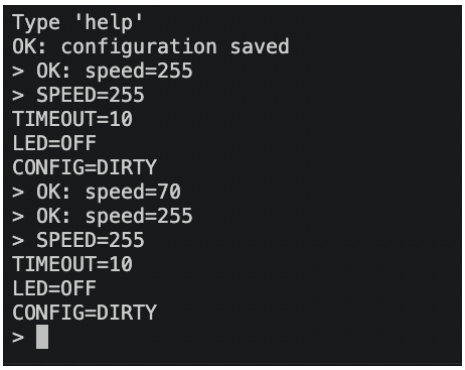
Expected: ‘ERROR: speed out of range’ in both states. Configuration unchanged; status remains available. REQ-02, REQ-03.

Actual: In both SAVED and DIRTY states, setting speed 256 sets speed to 255 instead of returning the ERROR. Status remains available

Severity: Major

Reproducibility: 5/5 Reproducible

Evidence:



BUG-02

Title: Timeout was set to 0 instead of returning an ERROR.

Steps:

1. Type command ‘save’
2. set timeout 0
3. Type command ‘status’
4. set timeout 10
5. set timeout 0
6. Type command ‘status’

Expected: ‘ERROR: timeout out of range’ in both states. Configuration unchanged; status remains available. REQ-04, REQ-05

Actual: In both SAVED and DIRTY states, setting timeout 0 sets timeout to 0 instead of returning the ERROR. Status remains available

Severity: Major

Reproducibility: 5/5 Reproducible

Evidence:

```
> OK: configuration saved
> OK: timeout=0
> SPEED=10
TIMEOUT=0
LED=OFF
CONFIG=DIRTY
> OK: timeout=10
> OK: timeout=0
> SPEED=10
TIMEOUT=0
LED=OFF
CONFIG=DIRTY
>
```

BUG-03

Title: CLI incorrectly accepts uppercase STATUS command

Steps: 1. Type 'STATUS'

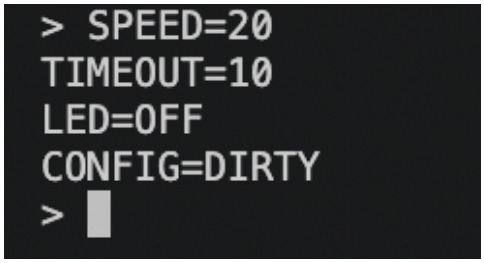
Expected: 'ERROR: unknown command' is displayed. REQ-06, REQ-07

Actual: Status is displayed

Severity: Minor

Reproducibility: 5/5 Reproducible

Evidence:



BUG-04

Title: System crashes when input exceeds the 40-character limit **(Related to BUG-06)**

Steps:

1. Type 'abcdefghijklmnopqrstuvwxyz123456789012345'

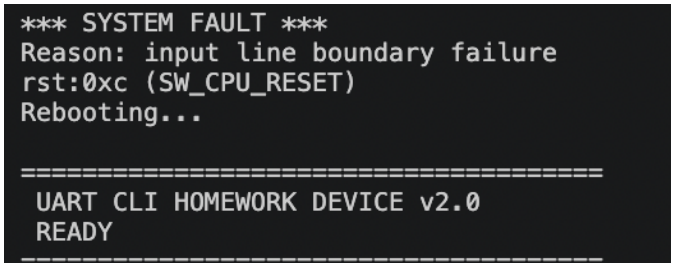
Expected: `ERROR: command too long`. REQ-10, REQ-11, REQ-12, REQ-16

Actual: System crashes and reboots. The ERROR is not displayed.

Severity: Critical

Reproducibility: 5/5 Reproducible

Evidence:



BUG-05

Title: Special characters and Unicode characters are incorrectly accepted as valid input

Steps:

1. Type '???' or 'так'

Expected: ERROR: unknown command. REQ-07, REQ-08

Actual: ERROR is not displayed. 'OK' returns

Severity: Major

Reproducibility: 5/5 Reproducible

Evidence:



BUG-06

Title: Crash. After reboot, CLI commands (help, status) return ERROR: unknown command **(Related to BUG-04)**

Steps:

1. Enter 'set speed 999999999999999999' 5 times consecutively without pressing Enter between inputs.
2. Press Enter once after the fifth input.

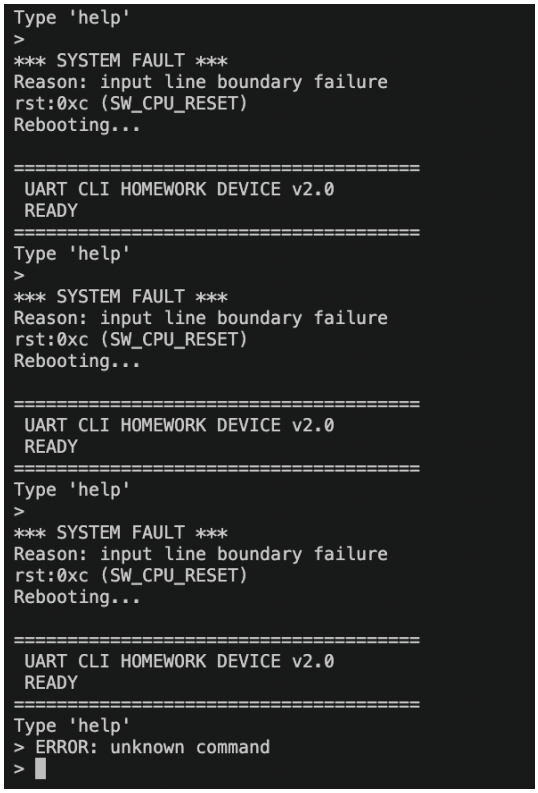
Expected: `ERROR: command too long`. Commands should respond. REQ-10, REQ-11, REQ-12, REQ-15, REQ-16

Actual: System crashes and reboots. The ERROR is not displayed. After typing 'help' or 'status' it returns ERROR: unknown command.

Severity: Critical

Reproducibility: 5/5 Reproducible

Evidence:



BUG-07

Title: Crash. The device reports ‘parser state mismatch while configuration is dirty’ and reboots after processing the invalid input sequence.

Steps:

Enter the commands one by one

1. set speed 50
2. save
3. set
4. status
5. set speed 70
6. set

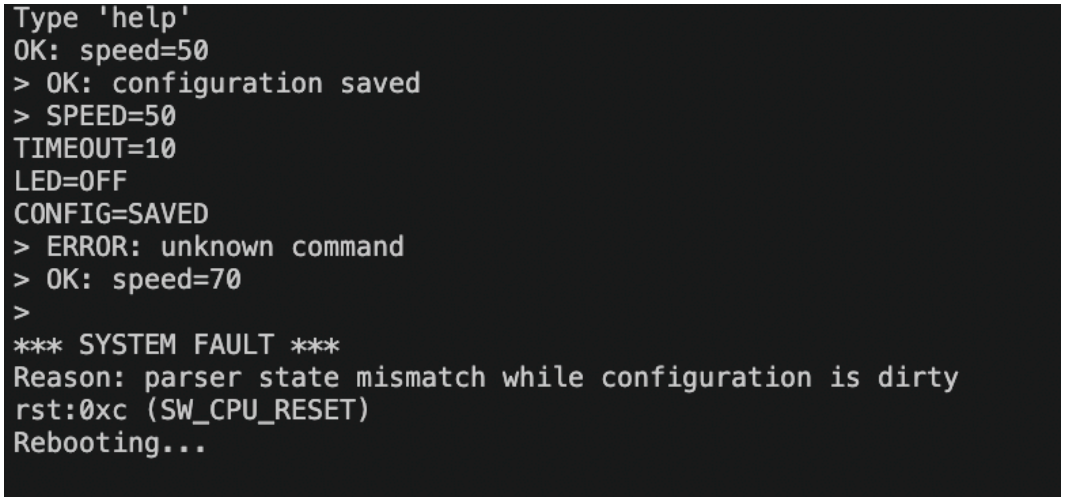
Expected: ERROR: unknown command for ‘set’ command; set speed 50/70 returns OK; speed=50/70; status is displayed. REQ-07, REQ-08, REQ-14, REQ-15, REQ-16

Actual: System crashes and reboots with parser state mismatch while configuration is dirty. The ERROR is not displayed. In SAVE state ‘ERROR: unknown command’ for ‘set’ command returns.

Severity: Critical

Reproducibility: 5/5 Reproducible

Evidence:



4. QA Conclusion: FAIL

UART CLI testing was performed using the following techniques: Boundary Value Analysis; Negative Testing; Input Length Testing; State / Sequence Testing; Stress Input; Functional Testing.

Boundary Value Analysis failed for both 'set speed' and 'set timeout'.
Input Length Testing failed because entering 41 characters causes the system to crash and reboot. In another case, after such a crash, the important 'status' command does not respond correctly.
Several bugs were observed during Negative Testing, State / Sequence Testing, and Stress Input testing. One more system crash was also observed after entering invalid commands. Several requirements also need clarification.

All identified bugs should be investigated and fixed.